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Fluid-Structure Interaction

Verification of a One-Way Coupled Fluid-Structure Interaction
Model with Small Solid Displacements via Manufactured
Solutions

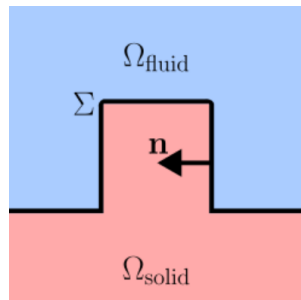
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Numerical Methods for Partial Differential Equations

January 23, 2026

Fluid–Structure Interaction Problem

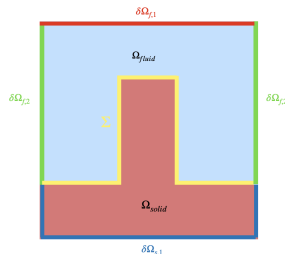
- **Steady, linear FSI problem**
- **Fluid:** Newtonian, incompressible
- **Solid:** linear, isotropic elastic
- **Domain decomposition:**
 - Fluid domain Ω_{fluid}
 - Solid domain Ω_{solid}
 - Interface Σ
- **Small displacement assumption**[1]:
 - No-slip condition
 - One-way coupling (Fluid $\rightarrow$ Solid)



Formulation and Analysis

The Problem: Steady Linear FSI

$$\left\{ \begin{array}{ll} -2\nu \nabla \cdot \varepsilon(\mathbf{u}) + \nabla p = 0, & \text{in } \Omega_{\text{fluid}}, \\ \nabla \cdot \mathbf{u} = 0, & \text{in } \Omega_{\text{fluid}}, \\ -\nabla \cdot \sigma(\mathbf{d}) = 0, & \text{in } \Omega_{\text{solid}}, \\ \mathbf{u} = \mathbf{0}, & \text{on } \Sigma = \partial\Omega_{\text{fluid}} \cap \partial\Omega_{\text{solid}}, \\ \sigma(\mathbf{d})\mathbf{n} = (2\nu \varepsilon(\mathbf{u}) - p\mathbf{I})\mathbf{n}, & \text{on } \Sigma, \\ \mathbf{u} = \mathbf{g}, & \text{on } \partial\Omega_{f,1}, \\ (2\nu \varepsilon(\mathbf{u}) - p\mathbf{I})\mathbf{n} = \mathbf{0}, & \text{on } \partial\Omega_{f,2}, \\ \mathbf{d} = \mathbf{0}, & \text{on } \partial\Omega_{s,1}. \end{array} \right.$$



Symmetric gradient definition:

$$\varepsilon(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$$

Stress tensor definition:

$$\sigma(\mathbf{d}) = 2\mu \varepsilon(\mathbf{d}) + \lambda \operatorname{tr}(\varepsilon(\mathbf{d})) \mathbf{I}$$

Weak Formulation_[2] of the Coupled FSI Problem

Functional spaces

$$\mathbf{V} := \{\mathbf{v} \in H^1(\Omega_f)^d : \mathbf{v} = \mathbf{g} \text{ on } \partial\Omega_{f,1}, \mathbf{v} = 0 \text{ on } \Sigma\},$$

$$\mathbf{V}_0 := \{\mathbf{v} \in H^1(\Omega_f)^d : \mathbf{v} = 0 \text{ on } \partial\Omega_{f,1} \cup \Sigma\},$$

$$Q := L^2(\Omega_f),$$

$$\mathbf{W} := \{\mathbf{w} \in H^1(\Omega_s)^d : \mathbf{w} = 0 \text{ on } \partial\Omega_{s,1}\}.$$

- **Fluid viscosity:** $a(\mathbf{u}, \mathbf{v}) := 2\nu \int_{\Omega_{\text{fluid}}} \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) d\Omega.$
- **Incompressibility:** $b(p, \mathbf{v}) := - \int_{\Omega_{\text{fluid}}} p \nabla \cdot \mathbf{v} d\Omega, \quad b(q, \mathbf{u}) := \int_{\Omega_{\text{fluid}}} q \nabla \cdot \mathbf{u} d\Omega.$
- **Fluid–solid interface:** $d((\mathbf{u}, p), \mathbf{b}) := - \int_{\Sigma} (2\nu \boldsymbol{\varepsilon}(\mathbf{u}) - p\mathbf{I})\mathbf{n} \cdot \mathbf{b} d\Gamma.$
- **Solid elasticity:** $s(\mathbf{d}, \mathbf{b}) := \int_{\Omega_{\text{solid}}} \boldsymbol{\sigma}(\mathbf{d}) : \boldsymbol{\varepsilon}(\mathbf{b}) d\Omega.$

$$a(\mathbf{u}, \mathbf{v}) + b(p, \mathbf{v}) + b(q, \mathbf{u}) + d((\mathbf{u}, p), \mathbf{b}) + s(\mathbf{d}, \mathbf{b}) = 0.$$

Dirichlet lifting

To handle non-homogeneous Dirichlet boundary conditions on $\delta\Omega_{f,1}$, we decompose the fluid velocity as

$$\mathbf{u} = \mathbf{u}_0 + \mathbf{u}_g,$$

where

$$\mathbf{u}_0 \in \mathbf{V}_0, \quad \mathbf{u}_0 = 0 \text{ on } \delta\Omega_{f,1} \cup \Sigma,$$

and $\mathbf{u}_g \in \mathbf{V}$ is a lifting of the prescribed boundary data $\mathbf{g}$, satisfying

$$\mathbf{u}_g = \mathbf{g} \quad \text{on } \delta\Omega_{f,1}.$$

Weak Formulation

find $(\mathbf{u}_0, p, \mathbf{d}) \in \mathbf{V}_0 \times Q \times \mathbf{W}$ such that

$$\begin{aligned} a(\mathbf{u}_0, \mathbf{a}) + b(p, \mathbf{a}) + b(q, \mathbf{u}_0) + d((\mathbf{u}_0, p); \mathbf{b}) + s(\mathbf{d}, \mathbf{b}) \\ = -a(\mathbf{u}_g, \mathbf{a}) - b(q, \mathbf{u}_g) - d((\mathbf{u}_g, 0), \mathbf{b}), \end{aligned}$$

$$\forall (\mathbf{a}, q, \mathbf{b}) \in \mathbf{V}_0 \times Q \times \mathbf{W}.$$

Galerkin Formulation and Discrete Matrices Representation

We introduce the finite-dimensional spaces

$$\mathbf{V}_h \subset \mathbf{V}_0, \quad Q_h \subset Q, \quad \mathbf{W}_h \subset \mathbf{W},$$

with bases $\{\phi_i^u\}_{i=1}^{N_u}, \{\psi_j\}_{j=1}^{N_p}, \{\phi_k^d\}_{k=1}^{N_d}$.

The discrete expansions are

$$\mathbf{u}_h = \sum_i U_i \phi_i^u, \quad p_h = \sum_j P_j \psi_j, \quad \mathbf{d}_h = \sum_k D_k \phi_k^d.$$

Each bilinear form induces an algebraic block:

- **Fluid viscosity:** $a(\mathbf{u}_h, \mathbf{v}_h) \longrightarrow \mathbf{A} \in \mathbb{R}^{N_u \times N_u}$
- **Incompressibility:** $b(p_h, \mathbf{v}_h), b(q_h, \mathbf{u}_h) \longrightarrow \mathbf{B}^T, \mathbf{B}$
- **Fluid–solid interface:** $d((\mathbf{u}_h, p_h), \mathbf{b}_h) \longrightarrow \mathbf{D}_u, \mathbf{D}_p$
- **Solid elasticity:** $s(\mathbf{d}_h, \mathbf{b}_h) \longrightarrow \mathbf{S} \in \mathbb{R}^{N_d \times N_d}$

Block Structure of the Discrete System

The discrete unknowns are

$$\mathbf{U} \in \mathbb{R}^{N_u}, \quad \mathbf{P} \in \mathbb{R}^{N_p}, \quad \mathbf{D} \in \mathbb{R}^{N_d}.$$

This yields the 3×3 **algebraic system**

$$\begin{bmatrix} \mathbf{A} & \mathbf{B}^T & \mathbf{0} \\ \mathbf{B} & \mathbf{0} & \mathbf{0} \\ \mathbf{D}_u & \mathbf{D}_p & \mathbf{S} \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \mathbf{P} \\ \mathbf{D} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_u \\ \mathbf{F}_p \\ \mathbf{0} \end{bmatrix}.$$

At the modeling level, the problem admits a 2×2 **fluid–solid structure**. The fluid operator is the Stokes block

$$\mathbf{A}_{\text{fluid}} = \begin{pmatrix} \mathbf{A} & \mathbf{B}^T \\ \mathbf{B} & \mathbf{0} \end{pmatrix},$$

while the solid operator is $\mathbf{S}$.

The resulting 2×2 **block form** is

$$\begin{pmatrix} \mathbf{A}_{\text{fluid}} & \mathbf{0} \\ (\mathbf{D}_u & \mathbf{D}_p) & \mathbf{S} \end{pmatrix},$$

with coupling induced by traction continuity on Σ .

Existence and Uniqueness: Coupled Problem

Stokes problem: find $(\mathbf{u}_h, p_h) \in \mathbf{V}_h \times \mathbf{Q}_h$ s.t.

$$\begin{cases} a(\mathbf{u}_h, \mathbf{v}_h) + b(\mathbf{v}_h, p_h) = a(\mathbf{u}_g, \mathbf{v}_h), \\ b(\mathbf{u}_h, q_h) = -b(q_h, \mathbf{u}_g) \end{cases}$$

Existence & uniqueness: Brezzi theorem.

Conclusion. The coupled Galerkin problem admits a *unique discrete solution* $(\mathbf{u}_h, p_h, \mathbf{d}_h)$.

Solid problem: find $\mathbf{d}_h \in \mathbf{W}_h$ s.t.

$$s(\mathbf{d}_h, \mathbf{b}_h) = d((\mathbf{u}_h, p_h), \mathbf{b}_h) = \int_{\Sigma} \mathbf{t}_h \cdot \mathbf{b}_h$$

$$\mathbf{t}_h = (2\nu\epsilon(\mathbf{u}_h) - p_h\mathbf{I})\mathbf{n}$$

Existence & uniqueness: Lax–Milgram.
($\mathbf{t}_h \in [H^{-1/2}(\Sigma)]^d$)

Analysis: Stability

Stability of the coupled problem follows from the block-wise analysis.

Fluid problem (Stokes).

$$\|\mathbf{u}_h\|_V + \|\mathbf{p}_h\|_Q \leq C_f \|f\|_{V'},$$

$C_f > 0$ independent of the mesh size h .

Solid problem (linear elasticity).

$$\|\mathbf{d}_h\|_V \leq \frac{1}{\alpha_s} \|d(\mathbf{u}_h, \mathbf{p}_h; \cdot)\|_{V'}.$$

Interface coupling. The fluid–solid interface term defines a bounded linear functional:

$$\|d(\mathbf{u}_h, \mathbf{p}_h; \cdot)\|_{V'} \leq C (\|\mathbf{u}_h\|_V + \|\mathbf{p}_h\|_Q).$$

Global stability. Combining the above estimates yields

$$\|\mathbf{u}_h\|_V + \|\mathbf{p}_h\|_Q + \|\mathbf{d}_h\|_V \leq C \|f\|_{V'},$$

with a constant $C > 0$ independent of h .

Analysis: Convergence

Fluid problem (Stokes)

Using Taylor–Hood elements $P_k/P(k-1)$, the discrete solution satisfies

$$\|\mathbf{u} - \mathbf{u}_h\|_V + \|p - p_h\|_Q \leq C h^k \left(\|\mathbf{u}\|_{H^{k+1}(\Omega_f)} + \|p\|_{H^k(\Omega_f)} \right).$$

Solid problem

By Céa's lemma, the displacement error satisfies

$$\|\mathbf{d} - \mathbf{d}_h\|_V \leq C \left(h^r \|\mathbf{d}\|_{H^{r+1}(\Omega_s)} + \|\mathbf{u} - \mathbf{u}_h\|_V + \|p - p_h\|_Q \right).$$

Combining both results yields

$$\|\mathbf{d} - \mathbf{d}_h\|_W \leq C(h^r + h^k), \quad \|\mathbf{d} - \mathbf{d}_h\|_W = \mathcal{O}\left(h^{\min(r,k)}\right).$$

Using a higher polynomial degree for the solid ($r > k$) does not improve the asymptotic convergence rate, which is limited by the accuracy of the fluid traction on the interface.

Implementation

Framework

- **Library:** Implemented using C++ and the `deal.II` library.
- **Parallelization:** Implemented using MPI and Trilinos. (PETSc also used to debug)

Extended Variables on a Single Global Domain

For assembly on a *single global triangulation*, we extend all fields by zero to Ω :

$$\mathbf{u} \equiv \mathbf{0} \text{ in } \Omega_s, \quad p \equiv 0 \text{ in } \Omega_s, \quad \mathbf{d} \equiv \mathbf{0} \text{ in } \Omega_f.$$

Extended functional spaces

$$\mathbf{V}^{\text{ext}} := \left\{ \mathbf{u} \in H^1(\Omega)^d \mid \mathbf{u} = \mathbf{0} \text{ in } \Omega_s, \mathbf{u} = \mathbf{g} \text{ on } \partial\Omega_{f,1}, \mathbf{u} = \mathbf{0} \text{ on } \Sigma \right\},$$

$$\mathbf{V}_0^{\text{ext}} := \left\{ \mathbf{v} \in H^1(\Omega)^d \mid \mathbf{v} = \mathbf{0} \text{ in } \Omega_s, \mathbf{v} = \mathbf{0} \text{ on } \partial\Omega_{f,1} \cup \Sigma \right\},$$

$$Q^{\text{ext}} := \left\{ p \in L^2(\Omega) \mid p = 0 \text{ in } \Omega_s \right\},$$

$$\mathbf{W}^{\text{ext}} := \left\{ \mathbf{d} \in H^1(\Omega)^d \mid \mathbf{d} = \mathbf{0} \text{ in } \Omega_f, \mathbf{d} = \mathbf{0} \text{ on } \partial\Omega_{s,1} \right\}.$$

Discrete Spaces via Zero Extensions

Let $\mathcal{T}_h$ be a triangulation of the global domain Ω . The discrete velocity and pressure spaces are defined by extending the Taylor–Hood pair by zero outside the fluid region:

$$\mathbf{V}_h = \left\{ \mathbf{u}_h \left| \begin{array}{l} \forall K \in \mathcal{T}_h : \mathbf{u}_h|_K \in \mathbb{P}_{p+1}^d \text{ if } K \subset \Omega_f, \\ \mathbf{u}_h|_{\Omega_f} \text{ continuous,} \\ \mathbf{u}_h|_K = \mathbf{0} \text{ if } K \subset \Omega_s \end{array} \right. \right\} \subset \mathbf{V}^{\text{ext}},$$

$$Q_h = \left\{ p_h \left| \begin{array}{l} \forall K \in \mathcal{T}_h : p_h|_K \in \mathbb{P}_p \text{ if } K \subset \Omega_f, \\ p_h|_{\Omega_f} \text{ continuous,} \\ p_h|_K = 0 \text{ if } K \subset \Omega_s \end{array} \right. \right\} \subset Q^{\text{ext}}.$$

The same construction is applied to the solid displacement using Lagrangian elements:

$$\mathbf{W}_h = \left\{ \mathbf{d}_h \left| \begin{array}{l} \forall K \in \mathcal{T}_h : \mathbf{d}_h|_K \in \mathbb{P}_r^d \text{ if } K \subset \Omega_s, \\ \mathbf{d}_h|_{\Omega_s} \text{ continuous,} \\ \mathbf{d}_h|_K = \mathbf{0} \text{ if } K \subset \Omega_f \end{array} \right. \right\} \subset \mathbf{W}^{\text{ext}}.$$

Global Discrete Solution Space

The discrete solution is the vector-valued field

$$y_h = (\mathbf{u}_h, p_h, \mathbf{d}_h),$$

belonging to the global space

$$Y_h = \left\{ y_h \left| \begin{array}{l} y_h|_{\Omega_f} \in \mathbb{P}_{p+1}^d \times \mathbb{P}_p \times Z^d, \\ y_h|_{\Omega_s} \in Z^d \times Z \times \mathbb{P}_r^d \end{array} \right. \right\}.$$

Here,

$$Z = \{\varphi : \varphi(x) = 0\}, \quad Z^d = \{\boldsymbol{\varphi} : \boldsymbol{\varphi}(x) = \mathbf{0}\},$$

denote the trivial scalar and vector-valued spaces used to represent fields that vanish identically on a subdomain.

hp-FEM Strategy

- **Multiple FE Systems:** We use `hp::FECollection` to handle distinct physical models (Stokes vs. Solid) within a single `DoFHandler`.
- **Zero Extension:** Variables are extended by zero in inactive subdomains using `FE_Nothing`, which contributes no degrees of freedom.
- **Material ID:** A cell-wise identifier selects the appropriate finite element (fluid or solid) and ensures correct refinement propagation.

Interface Assembly and Restricted Flux Coupling

The interface term is evaluated as a sum of integrals over interface faces:

$$- \sum_K (\psi_i[d], (2\nu\varepsilon(\psi_j[\mathbf{u}]) - \psi_j[p]I)\mathbf{n})_{\partial K \cap \Sigma}.$$

The quantities involved are defined on different sides of the interface:

- displacement test functions and normals on the solid side,
- velocity and pressure trial functions on the fluid side.

From an algebraic viewpoint, this term acts as a numerical flux.

Flux-based coupling is applied only on faces belonging to Σ , while standard cell-wise sparsity is retained elsewhere.

Adaptive Mesh Refinement

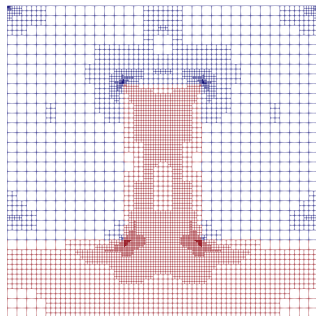
Solve–Estimate–Refine loop:

1. Solve the discrete problem on the current mesh.
2. Compute a posteriori error indicators (Kelly-type) per cell.
3. Refine cells with largest errors until tolerance or max level.

Error indicators: for a scalar field x_h on cell K ,

$$\eta_K^2 = \sum_{F \in \partial K} c_F \int_F [[\nabla x_h \cdot n]]^2 ds, \quad \eta^2 = \sum_K \eta_K^2.$$

Component-wise for vector fields; fluid and solid contributions are normalized before combination.



We consider two linear solver strategies for the monolithic fluid–structure system:

- **Direct solver (reference):** sparse decomposition using MUMPS, in sequential and parallel runs.
- **Iterative solver: FGMRES**, required by the use of *nonlinear and inexact block preconditioners*.

Block Preconditioner for the Coupled Problem

Exploiting the block-triangular structure, the inverse of the coupled operator can be written explicitly and applied sequentially.

The resulting **block preconditioner** is

$$\tilde{\mathbf{A}}_{\text{global}}^{-1} = \begin{pmatrix} \tilde{\mathbf{A}}_{\text{fluid}}^{-1} & \mathbf{0} \\ -\tilde{\mathbf{A}}_{\text{solid}}^{-1} \mathbf{B}_{\text{fs}} \tilde{\mathbf{A}}_{\text{fluid}}^{-1} & \tilde{\mathbf{A}}_{\text{solid}}^{-1} \end{pmatrix}.$$

The fluid block $\mathbf{A}_{\text{fluid}}$ is itself treated by a **nested block preconditioner** for the Stokes equations.

3×3 Block Preconditioner

The fully coupled system has a natural 3×3 block structure: fluid velocity, fluid pressure, solid displacement.

The global block preconditioner reads:

$$\tilde{\mathbf{A}}_{\text{global}}^{-1} = \begin{bmatrix} \mathbf{A}_{\text{stokes}}^{-1} & \mathbf{0} & \mathbf{0} \\ \mathbf{B} & \nu^{-1} \mathbf{M}_p^{-1} & \mathbf{0} \\ -\mathbf{P}_{\text{sol}}^{-1} \mathbf{D}_u \tilde{\mathbf{A}}_{\text{fluid}}^{-1} & -\mathbf{P}_{\text{sol}}^{-1} \mathbf{D}_p \tilde{\mathbf{A}}_{\text{fluid}}^{-1} & \mathbf{P}_{\text{sol}}^{-1} \end{bmatrix}.$$

Key points:

- Sequential application: velocity $\rightarrow$ pressure $\rightarrow$ solid
- Fluid block uses a nested Stokes preconditioner
- Solid block preconditioned with AMG

Sub-Preconditioners and Tuning

Fluid (Stokes) block:

$$\mathbf{A}_{\text{fluid}} = \begin{pmatrix} \mathbf{A}_{\text{stokes}} & \mathbf{B}^T \\ \mathbf{B} & \mathbf{0} \end{pmatrix}, \quad \tilde{\mathbf{A}}_{\text{fluid}}^{-1} = \begin{pmatrix} \mathbf{A}_{\text{stokes}}^{-1} & \mathbf{0} \\ \mathbf{B} & \nu^{-1} \mathbf{M}_p^{-1} \end{pmatrix},$$

Solid block: $\mathbf{A}_{\text{solid}}^{-1}$ approximated by AMG.

AMG setup (Trilinos ML):

- Constant modes provided for vector-valued operators;
- elliptic mode enabled;
- higher-order aggregation;
- Chebyshev smoother, 2 sweeps;
- aggregation threshold 0.02;
- W-cycle for the solid preconditioner.

Manufactured Solution: Construction Strategy

The manufactured solution is designed to verify convergence and accuracy of the coupled fluid–solid solver.

Key requirements:

- Fluid incompressibility in Ω_{fluid} .
- No-slip condition on the fluid–solid interface Σ .
- Satisfaction of interface coupling conditions.

Fluid velocity:

- Introduce a smooth stream function ψ .
- Define $\mathbf{u} = (\partial_y \psi, -\partial_x \psi)$ to ensure $\nabla \cdot \mathbf{u} = 0$.
- Multiply ψ by a function vanishing on Σ (squared) to enforce $\mathbf{u} = \mathbf{0}$ on the interface.

Manufactured Solution: Interface Coupling

Interface equilibrium: enforcing $\sigma(\mathbf{d}) \mathbf{n} = (2\nu \varepsilon(\mathbf{u}) - p\mathbf{I})\mathbf{n}$ is challenging due to discontinuous normals on Σ .

Construction approach:

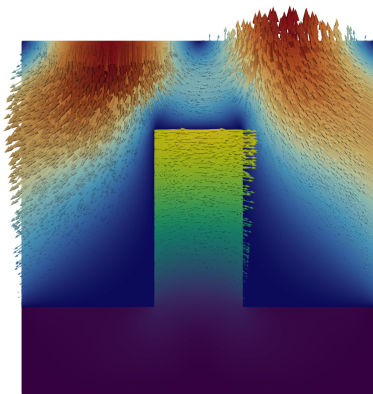
- Exploit the structural similarity of solid and fluid stress tensors.
- Choose $\mathbf{d} = \mathbf{c} + \mathbf{u}$ and consistent parameters (e.g. $\nu = \mu$).
- Define the pressure to match volumetric stress contributions.

This choice enforces equality of the stress tensors pointwise, making the interface condition satisfied by construction.

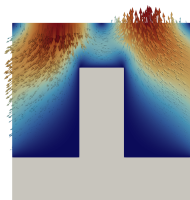
Forcing and boundary data:

- Forcing terms obtained by direct substitution.
- Dirichlet conditions set from the exact solution.
- Neumann conditions derived from the stress tensors.

Results



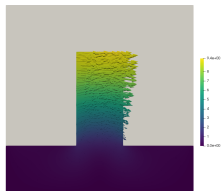
Numerical Results



Fluid velocity



Fluid pressure



Solid displacement (2D)

Hardware

- Single compute node
- Two Intel Xeon CPUs (28 cores each)
- 512 GB RAM

Post-processing

- Visualization in ParaView

Example in the images

- Uniform 128×128 mesh
- 421 573 DoFs
- Taylor–Hood $\mathbb{P}_2, \mathbb{P}_1$
- Linear elasticity $\mathbb{P}_1$

Robustness of the Block Preconditioner

Test setting

- 2D and 3D problems, up to $\mathcal{O}(10^6)$ DoFs
- Wide parameter ranges: $\nu \in [10^{-6}, 10^3]$, $\mu, \lambda \in [10^{-6}, 10^{10}]$
- Inner solvers: CG (10^{-2} relative tolerance)
- Outer solver: FGMRES (10^{-6} relative tolerance)

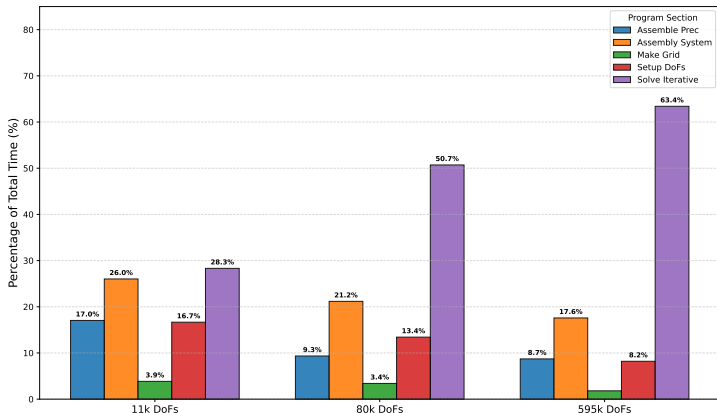
Observed iteration counts

	Velocity (CG)	Pressure (CG)	Elasticity (CG)	FGMRES
2D	10	2	10–20	20
3D	7	2	8	20

Key outcome

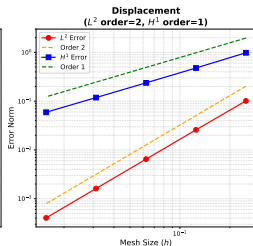
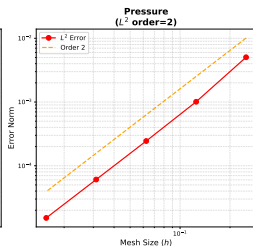
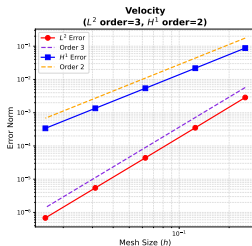
- Mesh-independent convergence for fluid and 3D elasticity blocks
- Mild growth only for 2D elasticity, reduced by multigrid W-cycles
- Overall robustness w.r.t. mesh refinement and parameter variation

Profiling

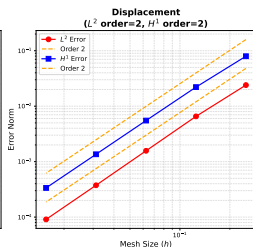
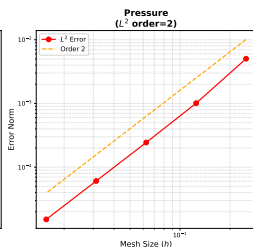
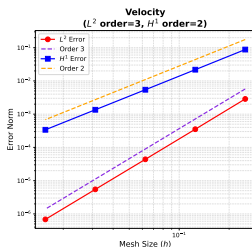


- Setup phases (mesh, DoFs) dominate only for small problems
- Iterative solve becomes the main cost as the problem size increases
- Assembly grows moderately with DoFs
- Preconditioner setup remains a minor fraction of total runtime

Convergence Analysis

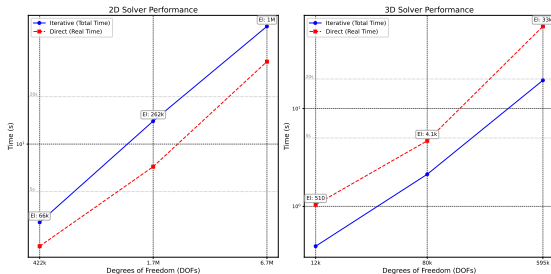


Taylor-Hood (P_2, P_1) for fluid, linear (P_1) elements for solid



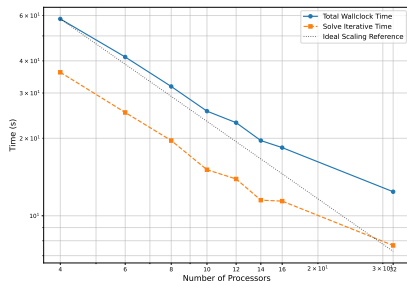
Taylor-Hood (P_2, P_1) for fluid, cubic (P_3) elements for solid

Iterative vs Direct Solvers

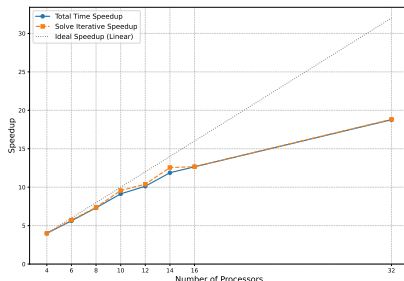


- 2D: direct solver faster due to limited fill-in and ample memory
- 3D: iterative solver outperforms direct, gap increases with problem size

Strong Scalability



Execution time vs. number of processors

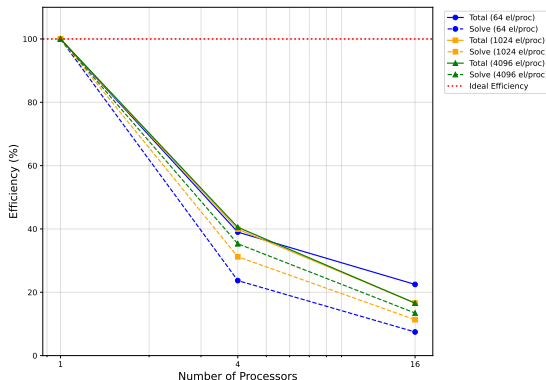


Parallel speedup vs. number of processors

Key observations:

- Nearly linear scaling up to 16 processors
- Saturation beyond 16 cores due to communication overhead

Weak Scalability



- Efficiency decreases as number of processors grows
- Iterative solver shows stronger degradation than total runtime
- Strong indication of communication-dominated scaling
- Study limited by Hardware and Domain Setup

Future Works

- **Extension to Navier-Stokes:** Incorporation of non-linear convective terms and temporal discretization. This step requires optimizing the assembly routines for both the system matrix and the preconditioner.
- **Moving Domains:** Development of a fully Arbitrary Lagrangian-Eulerian (ALE) framework.
- **Adaptive Mesh Refinement (AMR):** Investigation of alternative error estimators to drive the refinement process more effectively.
- **Solver Efficiency:** Transition to a matrix-free solver implementation to improve memory usage and scalability.

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Thank you for the attention!

